

Before Dawn

Made by Groupe I's members:

Florent Jin, Frédéric Lin
Kun Li and Daniel Abou Orm

December 1 2022

Introduction

Throughout this book of specifications, we will see many points of our project.

First, we will describe "Before Dawn", with a visual description of the game, explaining how the game is supposed to behave and become. Then we will explain its origin and why we decided that project.

Secondly, we will examine games that are similar to "Before Dawn". Nowadays, with the amount of out-comings games, almost every new games, including "Before Dawn", is inspired by old famous games. Though, the best blockbusters games are bringing new ideas, new concepts to the player. Thus, we will see how our game is bringing these ideas.

The last but not least, we will describe our project. First, let's see how the project will be done, thanks to which technological resources, that is to say, which tools, methods, software, and everything that is related too game development.

Then let's see how this project will upgrade ourselves, as individual persons and a group.

Finally, we will see how the project will be organized, first with the distribution of tasks for each member of the project, then with the progression throughout the time of the project, that means when a task should be at specific times.

In the end, a short conclusion will be done.

Contents

1	Who are we	3
1.1	Us	3
1.2	Florent	3
1.3	Daniel	3
1.4	Frédéric	4
1.5	Kun	4
2	What is “Before Dawn” ?	5
2.1	General Description	5
2.2	Origin	6
3	Compared to other	7
3.1	Historical Similar Games	7
3.2	Our Guideline	8
4	Before Dawn	9
4.1	Our Project	9
4.2	Humanly Speaking	9
4.2.1	Individual	9
4.2.2	As a group	10
4.3	Tasks	11
4.4	Organization	13
4.4.1	Distribution 1.0	13
4.4.2	Distribution 2.0	14
4.4.3	Progression	14
5	Economic aspects	15
5.1	Map	15
5.2	Characters	15
5.2.1	Design	15
5.2.2	Skills	15
5.3	Music	15
5.4	UI	15
5.5	How we make profit	16
6	Conclusion	17

1 Who are we

1.1 Us

As we all know nowadays, the majority of people are in front of their cell phones, their consoles, etc. Here in our group, we decided to give the world a new taste of gaming in which we would be working on marvelous graphics, and miraculous designs, which will grab the gamers attention. Let's not ignore that this project will be positively affecting us as group members, in many different aspects. It will be developing our communication skills, in which we will be talking for hours discussing the game and taking it on a very different high level every time we do, as well as maturing our problem-solving and patience skills due to the small and big problems we will be facing in the development of our game, which will be a humongous add to our resume in the future.

1.2 Florent

Hello ! My name is Florent JIN and I'm a first year EPITA student. From my childhood, I have been surrounded by many relatives who loved playing video games. Since I was young, most of the time, I was around watching my cousins play many games on Xbox or even on Wii. I got my first phone during my middle school, then I was able to play many mobile games and I find out what I liked and on the contrary what I didn't like. With all my experience in mobile games, I realized that I love to play Tower Defense. Thus with this project, I will be able to put all my knowledge acquired in programming to carry out this project.

1.3 Daniel

It's Daniel Abou-orm, an 18 years old international student which came from Lebanon to France to pursue his dreams in what he loves, and what he's ambitious about.

1.4 Frédéric

Hello! I am Frédéric, a freshman at EPITA. Even if it's my first year, I want to do something that I could be proud of myself when I'm older and say even if it's not perfect that it was worth trying.

1.5 Kun

Hello, I am Kun Li, a first-year student at EPITA. "Group I" team member. I think games have become more than just games, with the development of our society, games can be a way to provide people with stress relief, they can be e-sports, they can be a healing therapy, they can be all kinds of possibilities, they can be something that exists in human society, or they can be an idea that can't be realized in reality, and games are becoming more and more powerful and unique. So I want to make games.

2 What is “Before Dawn” ?

2.1 General Description

“Before Dawn” is a tower defense game. Tower defense (TD) is a subgenre of strategy games whose objective is to defend the player’s territory or property by hindering enemy attackers or preventing them from reaching exits, usually by placing defensive structures in their path of attack or along their path. This usually means building a variety of different structures that serve to automatically stop, hinder, attack, or destroy the enemy. Tower defense games are considered a subgenre of real-time strategy video games because of their real-time origins, although many modern tower defense games include aspects of turn-based strategy. The strategic selection and positioning of defensive elements is the basic strategy of the genre.

“Before Dawn” is a game that immerses the player in a futuristic and fantasy world. An evil force of darkness has invaded a beautiful world with an elf, fae, humans, and dwarfs who lived together and whenever darkness comes, they turn into monsters and invade the entire country. But a few people survived and to defend themselves, they decided to create robots to face monsters. You have been tasked with stopping the enemy’s advance, defending yourself from the invasion before the dawn comes. And your task is to set up an ambush on the path of the monsters or robots and prevent them from entering the camp.

When a monster or a robot is eliminated, the player is awarded bonuses or points to buy defensive buildings or upgrade them. Enemies attack in waves and most tower defense will pause after a wave to allow players to upgrade or add defensive buildings with points. There are approximately 10 waves or more in a game. Each headquarter has lives, and if the defensive structures fail to destroy the enemy, the enemy will have fewer headquarter lives when they reach a given place. As players upgrade their defense capabilities, the number and ability of monsters and robots will also improve. They improve faster than defensive buildings. So the goal of the game is to survive until the opponent has lost.

More than a game, “Before Dawn” builds a universe in which people can have fun as a single-player or as a multiplayer game. Players can choose different defensive structures to defend themselves against the werewolf invasion. Players can also choose their favorite heroes to walk on the map, and each has a very different skill set.

2.2 Origin

The idea of choosing a tower defense game was decided by our group through voting, discussion, name-calling, fighting and majority rule.

Thus, we wanted to make a game that allows the user to dream of futuristic fantasy.

Games as a product of software can bring surprise, beauty, joy, sadness and touch with screens and sound.

And as a human being, there may be nothing wrong with not having a dream and spending your life doing nothing, but flying hard for something beautiful is what we need from the bottom of our hearts.

So, we have fortunately decided firmly to make our own future for the pursuit of games that bring good experiences.

We will add magical effects, unique characters, and everything that fits into the futuristic and fantasy theme. The link between magic and games is the aspect of a dream. Indeed, playing video games is often about entering a universe created by someone. And there is nothing better than the use of magic in our dreams, and it can eventually remind people of their childhood memories.

3 Compared to other

3.1 Historical Similar Games

Our game is close to some famous games. Indeed, we can relate “Before Dawn” with “kingdom rush” for example. It uses the same Tower Defense principle. And “Kingdom rush” is the most popular game to introduce this game mode.

“Kingdom Rush” is a series of tower defense games set in a medieval fantasy setting. In line with the tower defense game genre, each level has a predetermined path where the player can build towers. Enemies will appear in waves on the path and the player’s task is to ensure that they are destroyed with towers before they reach the end of the road. By spending the gold gained from defeating enemies, the towers can be upgraded to deal more damage or, in the case of barracks, have more health. There are four basic types of towers to choose from in each match; mage, archer, barracks, and artillery. Different enemies in the game are resistant to certain types of damage, so which towers you place where is key to a good leveling strategy. Therefore, this game mode will be used in our gameplay.

There are also other similar games, for example, Bloons Tower Defense. The game is similar to Kingdom Rush, except that it has different levels of difficulty: easy, medium, and hard. As the player progresses through the maps, the variety of maps available becomes more varied. Some of the maps have pre-placed obstacles that may block the view of the monkey towers and their attacks. They also have multiplayer mode. We can also learn from this type of gameplay.

In multiplayer mode, we’ll take a leaf out of Clash Royale. It is a mobile game by the Finnish game company Supercell, a classic card strategy game where players play 1 vs 1 to find a random opponent, playing against each other with the cards they have, each with its own abilities. The game aims to destroy the opponent’s King’s Tower or to destroy the most Princess Towers before the game ends. We’ll take a page from this multiplayer approach and choose an attacker and a defender, and we’ll be adding a “Gacha game” element, where the heroes will be drawn at random.

3.2 Our Guideline

We are presenting these games because we were inspired by them. So to facilitate the presentation of our project, it will be a game with some cartoonish graphics. The gameplay is dynamic and the games are generally ten minutes long or even more, just like Kingdom Rush.

The choice of our project was the result of a long process of reflection: we first thought about which option was better between software and video games. Finally, we chose the video game option in a fair vote. For our project, we decided to write a tower defense program, which was the result of a vote including all different types of games.



Figure 1: Some photos from Kingdom Rush and Bloons TD

4 Before Dawn

Every developer has ever thought of making a game once in his life, and so do we.

4.1 Our Project

Our project will be simple. Before Dawn stands in a medieval and fantasy universe.

We chose this theme because this universe inspired us. Indeed, many things today take place in the fantasy world. For example, many animated series are from a fantasy genre, and we can quote the most famous, Attack on Titans. Thus, as we said before, thanks to the map that a friend gave us, we knew what should be done.

More than a basic school project, we want to implement good techniques and good features in Before Dawn. For example, we want our game to look good, and we'll do our best to succeed on this point.

4.2 Humanly Speaking

This project will improve our skills, whether it is individually or even by working in a group.

4.2.1 Individual

Firstly, individually, “Before Dawn” will make us learn many things. For example, it will improve our coding skills. Indeed, the best way to learn, even more in coding, is to experience, try, fail, and try again until finding a way to fix problems. Thus, by coding our project, we will learn many things on `c#` moreover, as the way of thinking while coding.

Then, this project will help us to manipulate software, more precisely in our case, Unity. Indeed, creating “Before Dawn” will teach us how to create a game. This will make it easier for us to create, or recreate, a game in the future. Furthermore, with the help of 2D software, for example, Blender, our team will learn how to design weapons, characters, effects, and all the stuff that needs to be created. For example, all the available characters in our games will be developed thanks to Blender (A bit for some, a lot for others).

To conclude on the learning part, throughout the end of the semester and the next semester, we will also learn how to make and implement audio on the game. Indeed, an audio part for the game will be developed. This will include hit sounds, music, and everything that is needed to immerse the player into our world.

Finally, this project will help us mentally. Indeed, with “Before Dawn”, we must take some responsibilities. With the project, we will need to behave well in order to finish the project. Each one of us needs to finish his tasks and need to take responsibility for what he should do on this project.

4.2.2 As a group

The S2 project is mainly a teamwork project because we all work together and split our work to be more efficient. So, this project can teach us how to manage our time as a group as much as how important it is to redistribute all tasks. Communication is an essential factor to carry out our project. As we come from different sections, we must overcome the language barrier by speaking english.

Making games gives us a better state of mind, logic, aesthetics, experience, sense of achievement, and a bigger picture. In fact, what we would like to do is of course to develop a game independently to our satisfaction. If we develop independently, the necessary planning and design management can exercise our big picture, while the bugs we encounter in the process of making the game and the difficulties we encounter in the learning process will allow us to help each other, asking questions on everything.

4.3 Tasks

Our team decided to offer the best experience for the player, whether it is visual or entertaining. But to reach our goal, this project will be divided into multiple categories in order to organize it.

- Gameplay : Every game has its foundation and no one will escape it. That's why we also need a foundation for our game, which is the playground where the player will fight each other, but also the process that leads to victory or even defeat. Therefore we will need to take care of the characteristics, like Health points, attacks, and heals of each character and each Hero. We must decide where the enemy will appear and how they disappear into the map.
- Graphics : To offer the best visual experience, we will have to take care of the design heroes and characters, with their weapons, their skills, and talents including animation of all characters. And not only that but also the display of the game which includes the field, game setting /setting of the game, Headquarters of races, and the interactions which are damages and shots from turrets.
- Multiplayer : This part is pretty "simple" to explain. Obviously, this is the creation of all the stuff that is needed in order to play as multiplayer.
- Artificial Intelligence (Enemy Pattern : The AI is used when we place our troops in map, they go through the map and when they encounter a tower, an enemy troop or a enemy Hero, they will rush into them.
- Artificial Intelligence (Enemy Pattern : The AI is also used by the turrets which must fire on enemies as soon as they see one and they are also within their range.
- UI : This is basically a user interface. It includes a menu, starting screen, settings, and everything that is in 2D.
- Sounds : This part will focus on finding/creating all the audio that we need to implement in the game. This is one of the most important parts, the audio part of the game immerse the player in the game.
- Gacha : We decided to add a new feature to give better entertainment to our players,

a system of chests that will allow players to unlock many heroes, which have their own history.

4.4 Organization

4.4.1 Distribution 1.0

For the organization, All the tasks have been divided between us: Each person in the group needs to be responsible for a small part, Even if the process is difficult, we will help each other.

4.4.1.1 Florent For my part, I decided to participate in the creation of the gameplay as a leader. I am also responsible for the multiplayer option with Daniel. Furthermore, I have the mission to support Kun for the sound, because I got some bases in music and also for the graphics with Frédéric because I always wanted to design characters.

4.4.1.2 Frederic During the project, group work, perseverance, attendance, and organization, will be essential for the achievement of our game. With my crazy ideas, I am responsible for creativity, which is why I am the leader of Graphics, and also of Gatcha, Kun and Florent will be attached with me because I need some feedback to know what could be good and what we need to improve. I also work on the UI, which permits me to improve my cooperation and organization for me as a future engineer.

4.4.1.2 Kun I love music, voice acting, so for my part, I want to take Sound part, and at the same time, I want to help Florent with the Gameplay, this is a big part that requires a lot of work and in which I can improve my understanding of the game. I want to help Daniel with the AI part, also, I hope I can help Frédéric with the Gacha part, writing their stories for the heroes of the game.

4.4.1.2 Daniel In this project, I will be responsible for the AI, and the UI, where I will be creating many user interfaces with the player such as the timer which can be turned on or off, and an option for the player to choose between a stealthy or aggressive mode where the AI will become more intelligent depending on what choice is made by the player. 12

4.4.2 Distribution 2.0

Tasks / Name	Daniel	Florent	Frederic	Kun
Gameplay				
Graphics				
Multiplayer				
UI-Game				
UI-Menu				
Sounds				
Gatcha				
AI-(E.P)				
AI-(T.H.P)				

E.P = Enemy Pattern

T.H.P = Tower/Heroes Pattern

 In charge

 Substitute

4.4.3 Progression

Tasks / Time	First presentation	Second presentation	Final presentation
Gameplay	20%	60%	100%
Graphics	10%	50%	100%
Multiplayer	0%	60%	100%
UI-Game	20%	50%	100%
UI-Menu	30%	50%	100%
Sounds	0%	50%	100%
Gatcha	40%	70%	100%
AI-(E.P)	30%	60%	100%
AI-(T.H.P)	30%	60%	100%

5 Economic aspects

5.1 Map

For the map, we are going to start from scratch, and go ahead step by step, maybe by getting some help from some objects already done online, but the idea is to start from zero, and gain our designing skills more and more as we advance in designing the map. Thus, there will not be any amount reserved for the map.

5.2 Characters

5.2.1 Design

We will allow the use of free online assets, but we will attempt to develop as many of our own characters as possible. By doing so, we reduce our working time on other aspects of the project that can be handled, but we also learn new skills and techniques that will be valuable and useful in the future.

5.2.2 Skills

The skills of the characters we construct will be developed using free materials and assets found online, adjusted, and we are going to be advancing in their level by level. So, instead of commencing from scratch for our concepts, we will be reaching for the assistance online and progressing in it further.

5.3 Music

For the musical part, we intended to use exclusively songs linked to defensive topics, as well as some enthusiastic sounds we could extract from music and sound effects.

5.4 UI

For the UI part, it will be done thanks to the old knowledge of things related to USER-INTERFACE done in the past, we will import some designs online, and edit them completely to fit into our game, and be compatible with what we are doing.

5.5 How we make profit

There are two ways that can be used to make a profit. We can choose to sell the game so the earnings will come from people who buy it or we can make the profit by selling the game without advertisements. The approach from the second one will not only save us from paying for advertisement and from having to pay for the amount of played time, but it can also make the players really enjoy playing our game without being disturbed. They will be able to focus on and learn how to play the game properly with no other thoughts. For example, if they want to get and hold a high score. As well as we can put some in-game purchases like skins, and coins or a game pass which will give a lot rewards for the player.

6 Conclusion

To sum up this specifications book, everybody is enthusiastic about this project because all of us have played video games in our lives. We hope to get it finalized by the end of the semester. We are fully conscious that there will be some difficulties related to specific aspects of the game, but we're going to do our best to succeed, which is the reason we chose to do each feature with a leader, and a substitute for it. Working together in groups can help you come up with fresh ideas and concepts as well as solutions as the game progresses. When we finish this project, we hope that it will improve our skills and techniques as well as our awareness of the importance of teamwork. Finally, we should indeed consider the economic aspect of the game because, although it is a project, it has the potential to become a real game with an economic aspect.